

Modelling a Modelling Community: Mapping and Simulating Agent-Based Research in Behavioural Social Science

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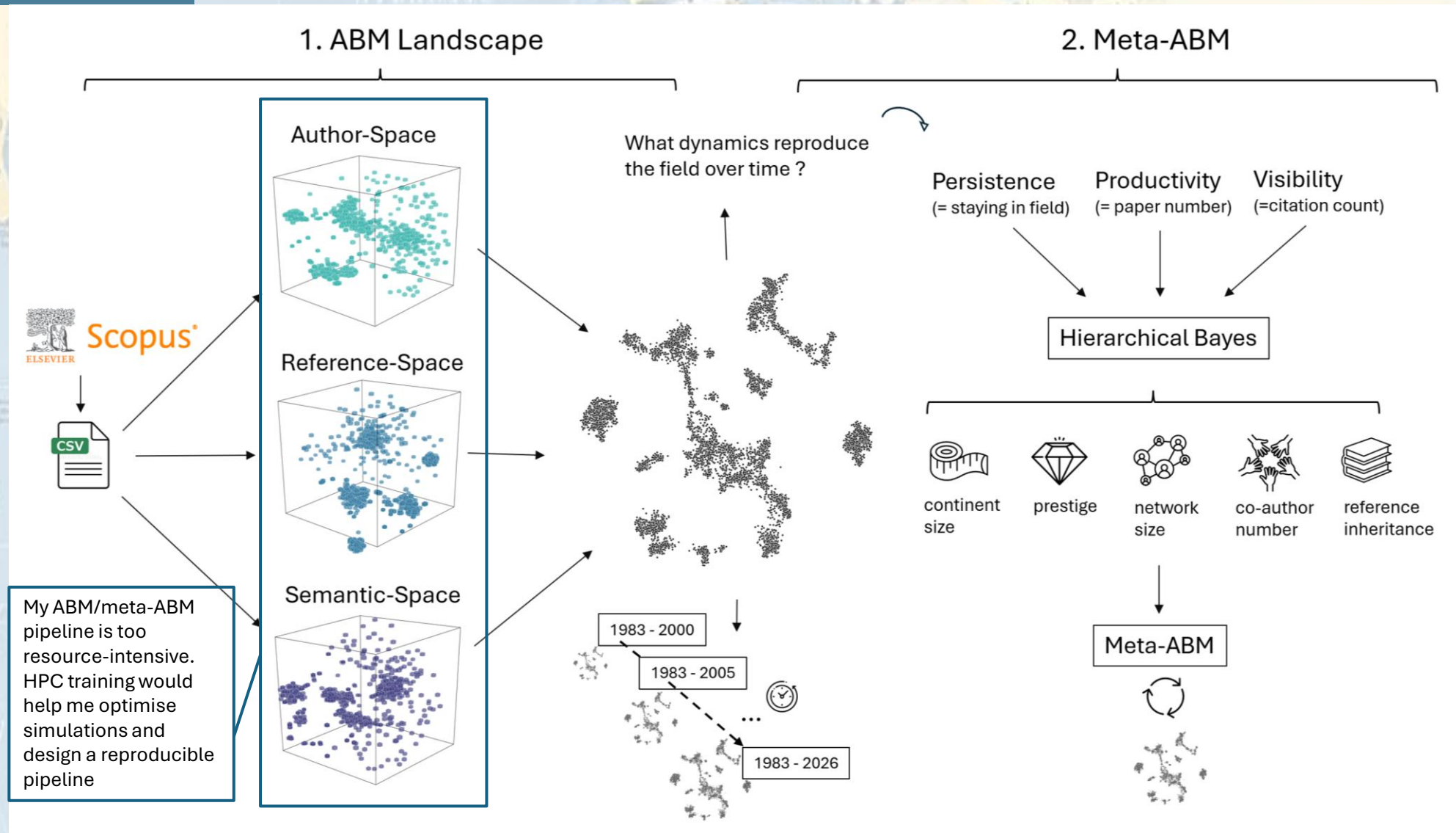
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Background

- We lack a systematic map of how **Agent-based Modelling (ABM)** research on human behaviour is organized across domains and how its subfields connect or diverge
- Existing reviews commonly focus on specific applications or narrow methodological strands^{2,3,4}
 1. To address this gap, we use an LLM-based bibliometric framework¹ (Thoma et al., 2025), to combine semantic, authorship, and reference patterns and map the intellectual landscape of ABM research across the behavioural and social sciences from 1983 to 2026 → **ABM landscape**
 2. Temporal Bayesian models test **what drives participation** (remaining in the field), **productivity** (number of papers), and **visibility** (number of citations) by testing ‘continent’ prestige, collaboration, team size, and reference reuse as predictors

Design



ABM Map

Combined									
Electricity Markets & Energy Transition	.43	.28	.45	.43	.44	.49	.47	.44	.73
Strategic Markets & Supply Chains	.00	.29	.53	.50	.51	.42	.48	1.00	.44
ABM Methodology	.47	.36	.59	.49	.57	.53	.72	.48	.47
Socio-Ecological Land & Water Systems	.41	.40	.50	.33	.44	.66	.53	.42	.49
Macroeconomic Dynamics & Collective Behaviour	.49	.34	.63	.65	.64	.44	.57	.51	.44
Financial Market Microstructure	.48	.24	.65	.95	.65	.33	.49	.50	.43
Social Diffusion & Cultural Evolution	.51	.42	.79	.65	.63	.50	.59	.53	.45
Epidemic & Pandemic Modeling	.30	.87	.42	.24	.34	.40	.36	.29	.28
Socioeconomic Infrastructure	1.00	.30	.51	.48	.49	.41	.47	.00	.43

Socioeconomic Infrastructure
Epidemic & Pandemic Modeling
Social Diffusion & Cultural Evolution
Financial Market Microstructure
Socio-Ecological Land & Water Systems
ABM Methodology
Strategic Markets & Supply Chains
Electricity Markets & Energy Transition

Author									
Electricity Markets & Energy Transition	.27	.15	.15	.22	.18	.21	.20	.30	.20
Strategic Markets & Supply Chains	.00	.26	.24	.32	.29	.31	.35	1.00	.30
ABM Methodology	.30	.16	.15	.22	.18	.21	.21	.35	.20
Socio-Ecological Land & Water Systems	.28	.16	.15	.23	.19	.22	.21	.31	.21
Macroeconomic Dynamics & Collective Behaviour	.28	.13	.13	.19	.16	.19	.18	.29	.18
Financial Market Microstructure	.41	.16	.16	.23	.19	.23	.22	.32	.22
Social Diffusion & Cultural Evolution	.20	.11	.11	.16	.13	.15	.15	.24	.15
Epidemic & Pandemic Modeling	.24	.12	.11	.16	.13	.16	.16	.26	.15
Socioeconomic Infrastructure	1.00	.24	.20	.41	.28	.28	.30	.00	.27

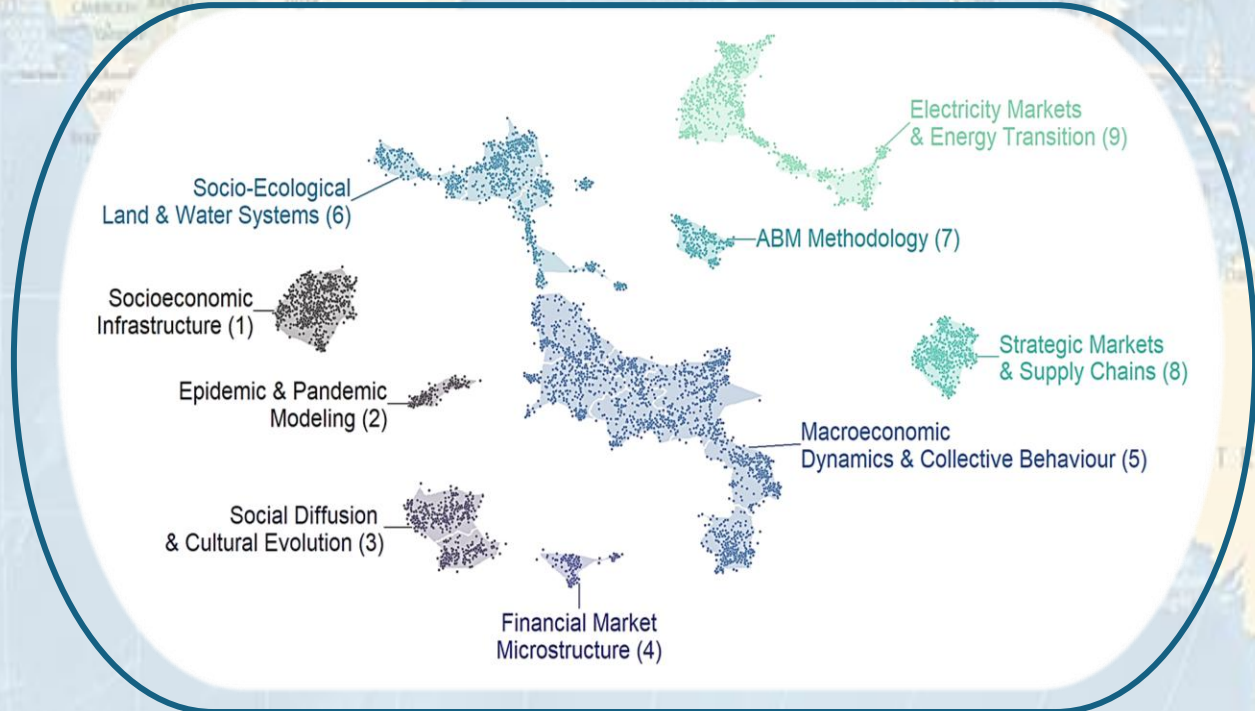
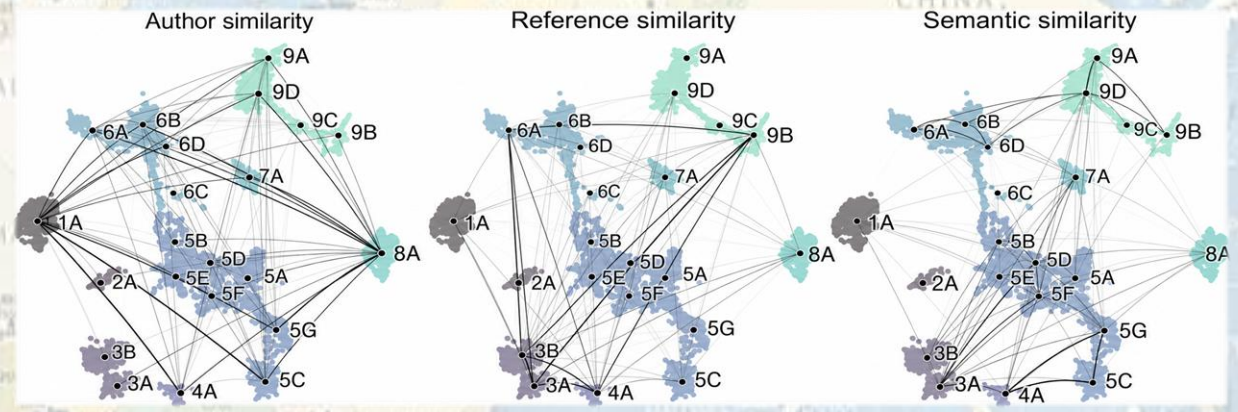
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Reference									
Electricity Markets & Energy Transition	.44	.41	.49	.48	.42	.45	.46	.45	.46
Strategic Markets & Supply Chains	.45	.43	.52	.50	.44	.46	.48	.46	.45
ABM Methodology	.48	.45	.49	.50	.46	.48	.71	.48	.46
Socio-Ecological Land & Water Systems	.45	.43	.51	.49	.43	.48	.48	.46	.45
Macroeconomic Dynamics & Collective Behaviour	.43	.41	.46	.46	.43	.43	.46	.44	.42
Financial Market Microstructure	.49	.45	.64	.76	.46	.49	.50	.50	.48
Social Diffusion & Cultural Evolution	.51	.46	.77	.64	.46	.51	.49	.52	.49
Epidemic & Pandemic Modeling	.42	.52	.46	.45	.41	.43	.45	.43	.41
Socioeconomic Infrastructure	.45	.42	.51	.49	.43	.45	.48	.45	.44

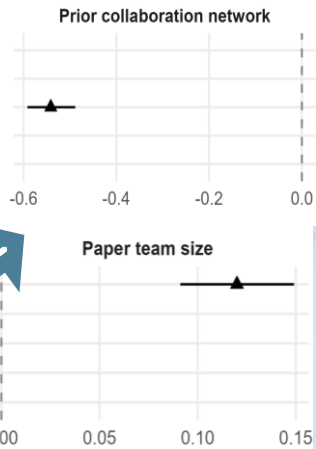
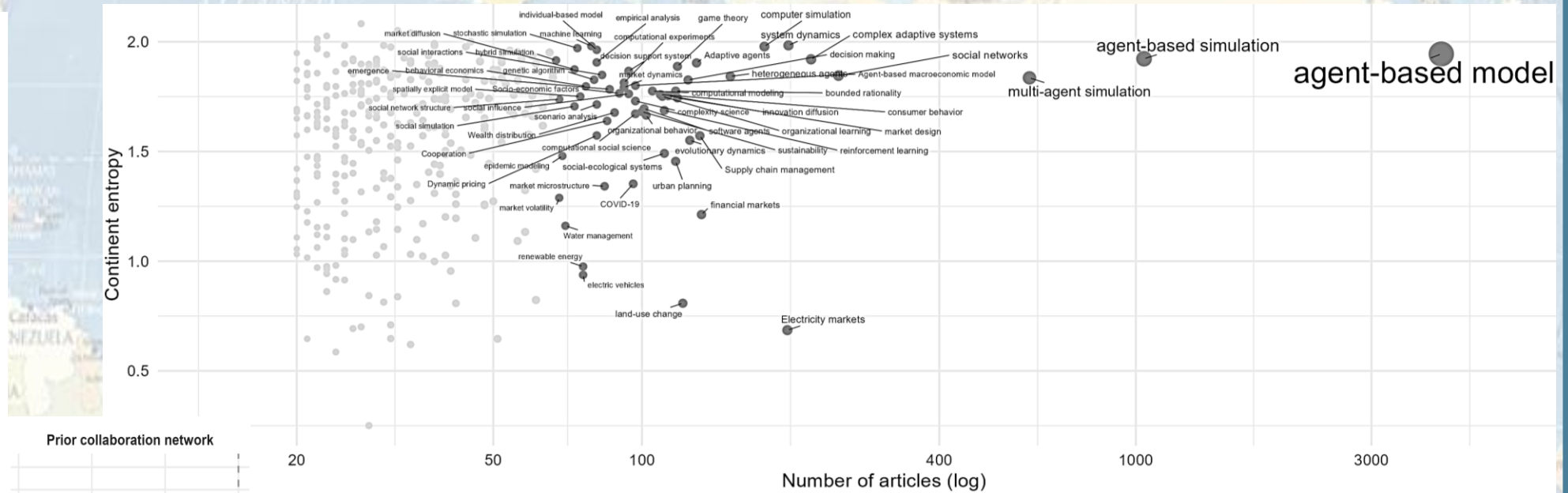
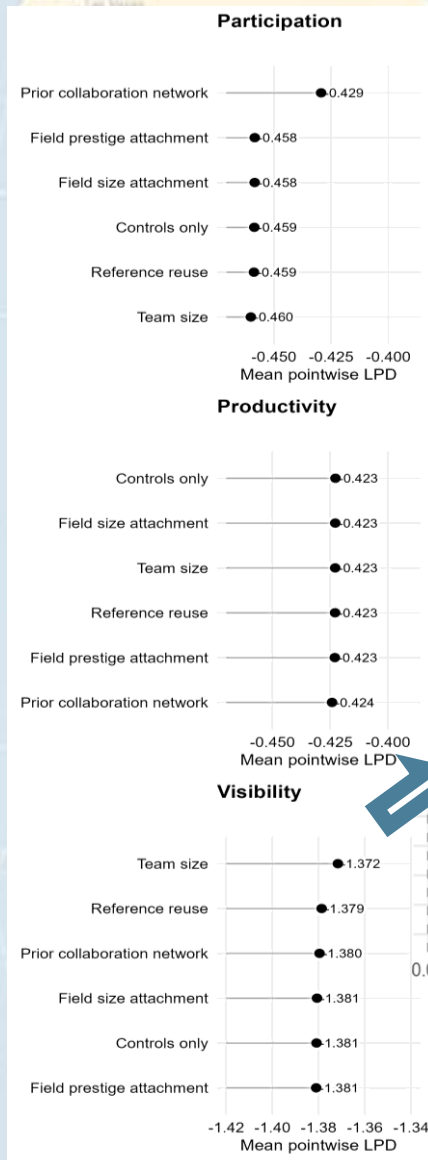
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Semantic									
Electricity Markets & Energy Transition	.47	.22	.40	.41	.41	.48	.48	.49	.72
Strategic Markets & Supply Chains	.50	.29	.54	.55	.55	.47	.55	.52	.49
ABM Methodology	.54	.34	.59	.52	.57	.58	.62	.55	.48
Socio-Ecological Land & Water Systems	.46	.35	.46	.32	.42	.66	.56	.47	.48
Macroeconomic Dynamics & Collective Behaviour	.52	.27	.60	.66	.62	.42	.57	.55	.41
Financial Market Microstructure	.52	.19	.54	.93	.66	.32	.52	.55	.41
Social Diffusion & Cultural Evolution	.52	.35	.64	.54	.60	.46	.59	.54	.40
Epidemic & Pandemic Modeling	.29	.84	.35	.19	.27	.35	.34	.29	.22
Socioeconomic Infrastructure	.49	.29	.52	.52	.52	.46	.54	.50	.47

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Scientific Merit + Tag Prevalence + Conclusion



- Behavioural ABM forms clear thematic regions, but these regions are only weakly integrated across authors, references, and institutions
- New ABM areas emerge as societal issues gain importance, such as energy transition and pandemic modelling
- Papers with more co-authors tend to be more visible (receive more citations) in behavioural ABM → even in a field that studies social systems, the production of science itself is socially organized
- LLM-assisted classification, embeddings, and expert interpretation can help synthesize large research areas more efficiently than traditional review methods alone

1. Thoma, A., Bolenz, F., Tiede, K., Yang, Y., Palminteri, S., Hertwig, R., & Wulff, D. (2025). Mapping the landscape of behavioral reinforcement learning research. https://osf.io/preprints/psyarxiv/6c2va_v1
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3. Conte, R., & Paolucci, M. (2014). On agent-based modeling and computational social science. *Frontiers in psychology*, 5, 668. <https://doi.org/10.3389/fpsyg.2014.00668>
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